

Usability and Design Principles

Usability Goals

User Experience Goals

Generic Design Principles

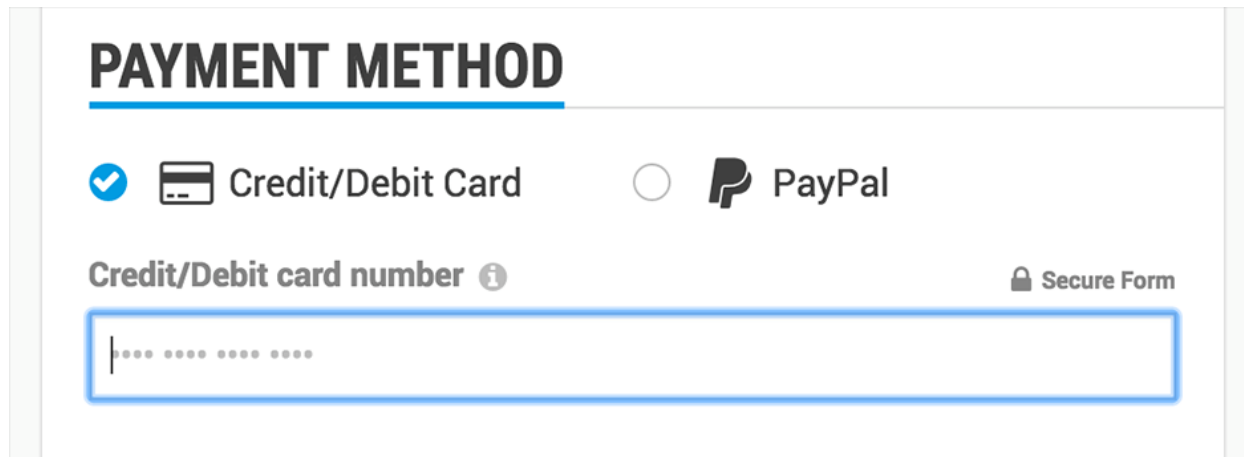
Usability

- Usability is a quality attribute that assesses how easy user interfaces are to use
- Some usability goals include:
 - Effectiveness
 - Efficiency
 - Safety
 - Utility
 - Learnability
 - Memorability

<https://www.nngroup.com/articles/usability-101-introduction-to-usability/>

Usability Goals

- Effectiveness
 - Is about whether users can complete their goals with the system
 - Question: is the system doing what it is supposed to do?
e.g. payment system, word processors, e-commerce system



The screenshot shows a web form titled "PAYMENT METHOD" in bold, underlined text. Below the title, there are two radio button options. The first option, "Credit/Debit Card", is selected, indicated by a blue checkmark in a circle next to a credit card icon. The second option, "PayPal", is unselected, indicated by an empty circle next to the PayPal logo. Below these options, there is a text input field for the "Credit/Debit card number". The field is currently empty, showing only a vertical cursor and a series of dots representing masked digits. To the right of the input field, there is a lock icon and the text "Secure Form".

Usability Goals

- Efficiency
 - Effective, ok; but efficient?
 - Considers how much time/number of steps it will take users to perform their tasks
 - Systems that require the least time/number of steps are usually considered as being efficient
 - Criteria: time to complete a task; # of operations to complete a task

Example

Student Service Centre
Course Schedule

Browse ▾ Search ▾ Help ▾

CWL Login

Course Schedule / Welcome to Course Schedule

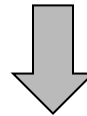
Campus: UBC Vancouver ▾ Session: 2018 Winter ▾

The University of British Columbia - Course Schedule

Welcome to UBC's Course Schedule. On this site you will be able to browse courses, save them to a worklist, view them in a timetable and even register for them!

This page requires that you have **Javascript** enabled on your browser. To enable Javascript, [click here](#).

Please note that the Course Schedule is **subject to change**. Please review your worklists prior to your registration date.



Student Service Centre
Course Schedule

Browse ▾ Search ▾ Help ▾

CWL Login

Course Schedule / Welcome to Course Schedule

Campus: UBC Okanagan ▾ Session: 2018 Winter ▾

The University of British Columbia - Course Schedule

Welcome to UBC's Course Schedule. On this site you will be able to browse courses, save them to a worklist, view them in a timetable and even register for them!

This page requires that you have **Javascript** enabled on your browser. To enable Javascript, [click here](#).

Please note that the Course Schedule is **subject to change**. Please review your worklists prior to your registration date.



Example

Student Service Centre
Course Schedule

Browse ▾

Search ▾

Help ▾

Courses

Instructor

Campus: UBC Okanagan ▾

Session: 2018 Winter ▾

CWL Login

Course Schedule

The University of British Columbia - Course Schedule

Welcome to UBC's Course Schedule. On this site you will be able to browse courses, save them to a worklist, view them in a timetable and even register for them!

This page requires that you have **Javascript** enabled on your browser. To enable Javascript, [click here](#).

Please note that the Course Schedule is **subject to change**. Please review your worklists prior to your registration date.

Registration Reminder: One-hour courses are actually only 50 minutes long and one-and-a-half-hour classes are 80 minutes long so you have time to get to your next class

Student Service Centre
Course Schedule

Browse ▾

Search ▾

Help ▾

CWL Login

Course Schedule / Search for Courses

Campus: UBC Okanagan ▾

Session: 2018 Winter ▾

Course Search: This is where you can search for courses based on a number of criteria including day and time offered, subject area and keywords in the course description. The results will display at the bottom of the page.

Search Criteria

Subject Area:

eg. ENGL or EN*

(eg. ENGL or EN*)

Course Number:

eg. 110 or 1*

(eg. 110 or 1*)

Keyword(s):

Results per Page

☒ 20 Items

☐ 50 Items

☐ 100 Items

Example

? Subject Area: (eg. EN*) 

? Course Number: (eg. 1) 

? Keyword(s):

? Activity Type:

? Credits:

? Term:

? Status:

? Available Times: - (24 Hr format)

? Available Days: ☐ Su ☐ M ☐ T ☐ W ☐ Th ☐ F ☐ Sa

? Check here if Available days must match precisely ☐

? Return Distance Education courses only ☐

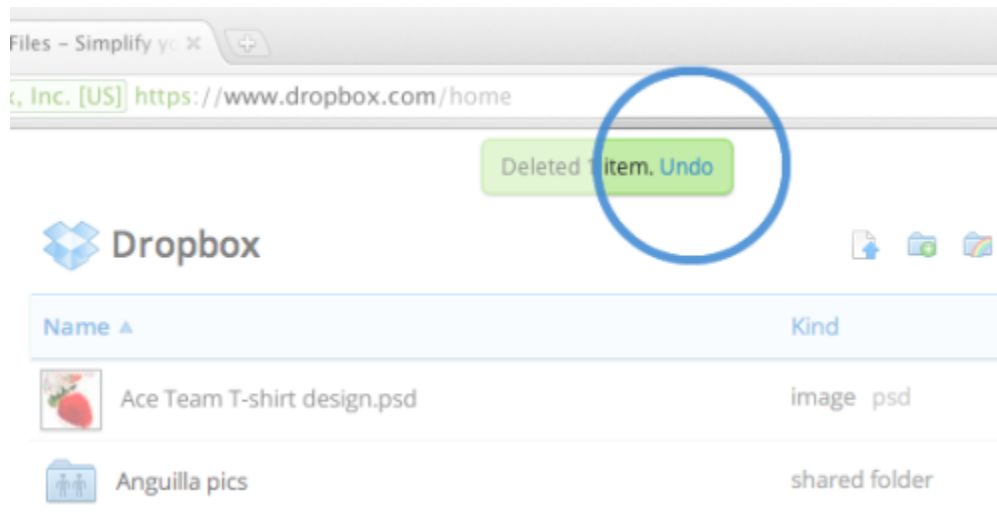
[Search for Sections](#) 

- ☐ 50 Items
- ☐ 100 Items

Status	Section	Activity	Term	Mode of Delivery	Interval	Days	Start Time	End Time	Section Comments	Course Requires In-Person Attendance
Full	COSC 341 101	Lecture	1	In-Person		Fri	17:00	20:00	 Section Comments	Yes
Full	COSC 341 L2A	Laboratory	1	In-Person		Wed	16:00	18:00		Yes
Full	COSC 341 L2B	Laboratory	1	In-Person		Mon	16:00	18:00		Yes
Full	COSC 341 L2C	Laboratory	1	In-Person		Wed	14:00	16:00		Yes
Full	COSC 341 L2D	Laboratory	1	In-Person		Fri	14:00	16:00		Yes
Full	COSC 341 L2E	Laboratory	1	In-Person		Mon	12:00	14:00		Yes
Full	COSC 341 L2F	Laboratory	1	In-Person		Wed	10:00	12:00		Yes

Usability Goals

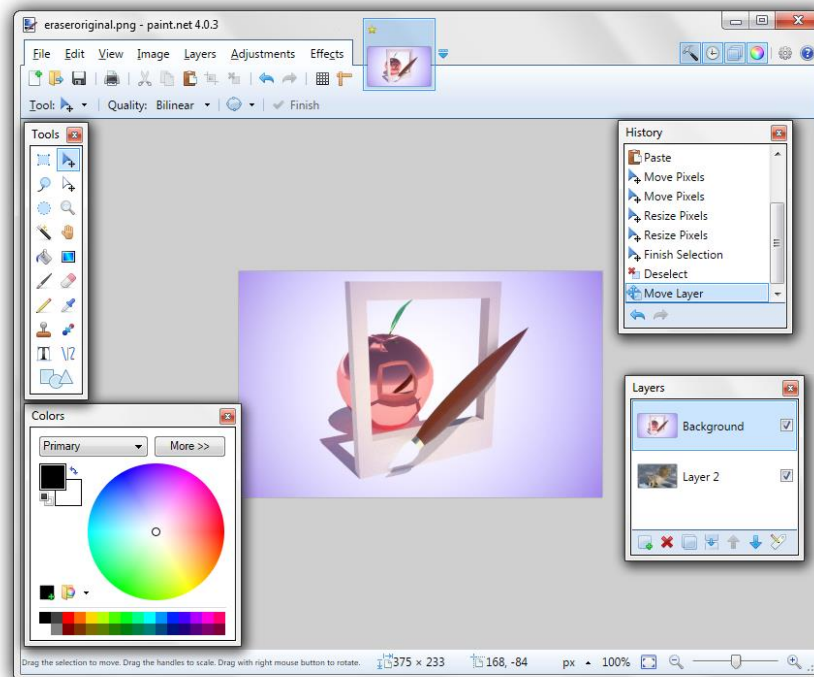
- Safety
 - prevent making serious/unrecoverable errors
 - provide means of recovering from errors
 - undo options, confirmation dialogs
 - Question: does it prevent users from making/recovering from serious errors?
 - Criteria: number of errors/time to recover from error
 - e.g., Dropbox



Usability Goals

- Utility

- sufficient functionality to accommodate range of users tasks
- Question: does it provide sufficient functionality for users to carry out tasks as naturally as possible?
- Criteria: availability of core tasks
 - e.g. accounting packages, drawing tools

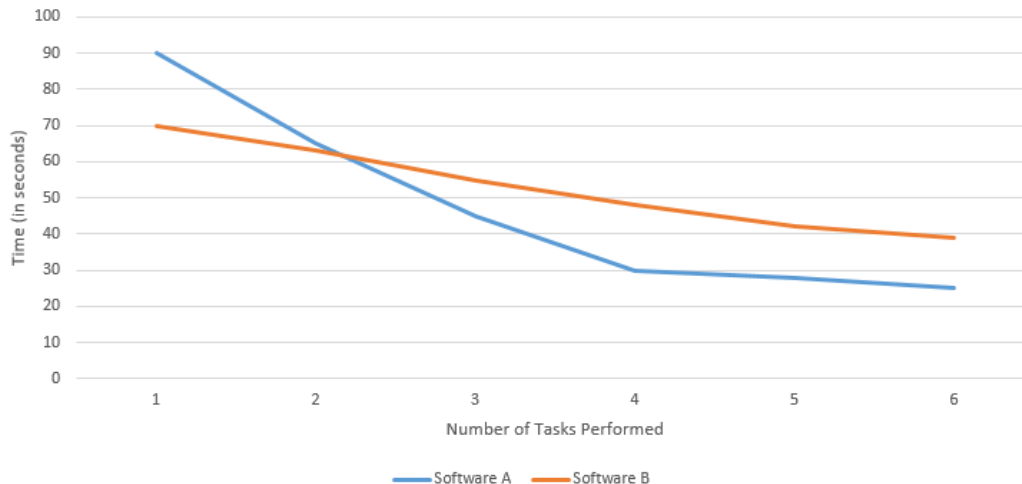


Usability Goals

▪ Learnability

- important if system will be adopted by the user
- A more learnable system is one that reduces the time it takes to complete tasks as users spend more time with a system faster than others
- Question: can primary (core) and secondary tasks be learned quickly and easily?
- Criteria: time to learn a task, errors made in learning a task
e.g. smartphone?

Learnability Comparison



Save



Search



Menu



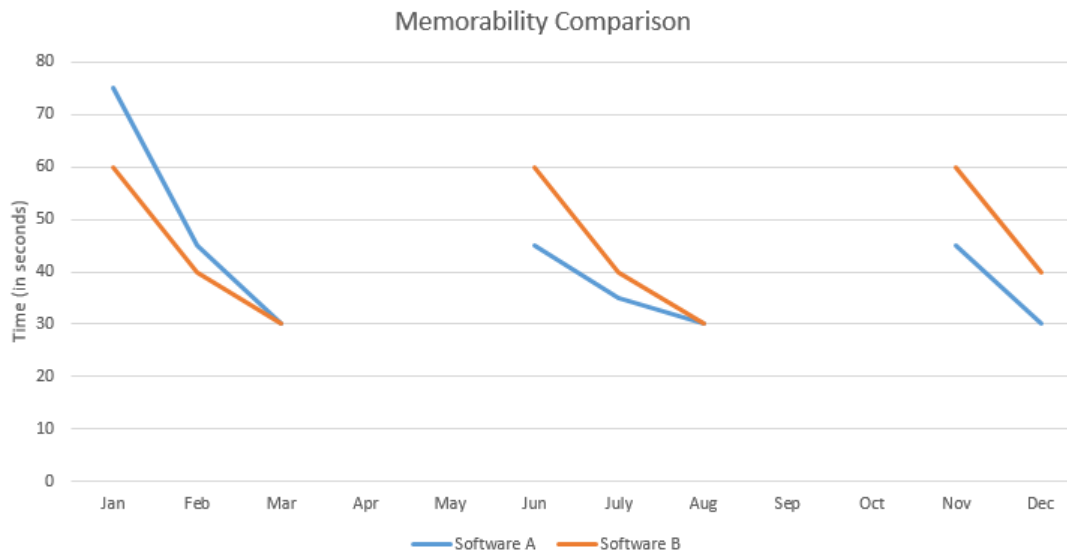
Basket



Call

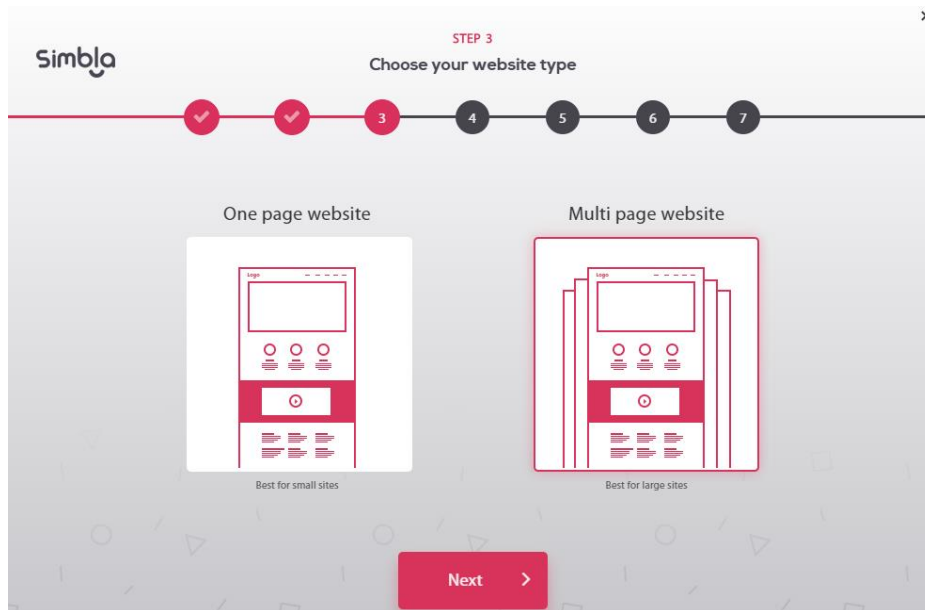
Usability Goals

- Memorability
 - concerned with how easy to remember once learned
 - important goal to include if systems will be used infrequently
 - Should the designer include meaningful icons, menu items, etc...
 - Question: will users remember all the steps used for carrying out a task?
 - Criteria: time or errors made in carrying out a task after system is learned



Example

- How long should it take to learn to use the following products?
How long does it actually take? How memorable are they?
- Using an authoring tool to create a website
- Setting up a wireless router



Example

- A user should be able to produce a basic web site in 20-30 mins. Most packages do achieve this goal (e.g., allowing the user publish content in web format). Some more complex operations will take longer, but users can learn and remember basic functionality relatively easily.
- Setting up a wireless router or access point is difficult to learn. It typically requires completing many steps, with terminology that is unfamiliar to most users. Many users are not able to do this at all and have to call for help from the “Geek Squad”. If its memorable, its because the user really wants to learn how to do it.

User Experience Goals

- User experience goals are concerned with the quality of the user's experience with the system
- They should enhance the user's experience

User Experience Goals

- A set of user experience goals include:
 - Aesthetically pleasing: subjective
 - Satisfying: productive
 - Motivating: did not feel like giving up
 - Enjoyable: no frustrations were encountered
 - Fun: excited about using it again
 - Supportive of creativity: drawing tools
 - Entertaining: games
 - Rewarding: sense of productivity
 - Helpful: clueless but still made it through

User Experience vs. Usability

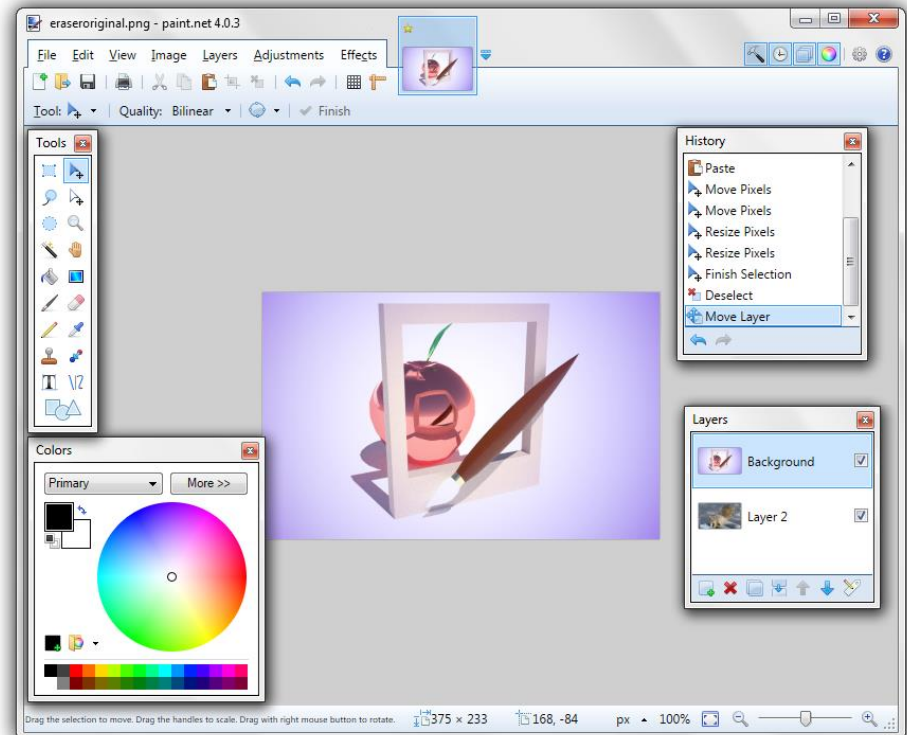
- Usability goals could take primary importance but not always
- User experience goals are not easily measured
- In some cases less-usable systems increase the user experience, indirect relationship (nintendo power pads for racing vs. hand pads/joysticks)
- Have to balance the tradeoffs between user experience and usability goals, i.e. be selective



Nintendo power pads: <https://www.youtube.com/watch?v=JjS71YRCBw4>

Discussion

- What are the key usability and user experience goals of these systems:
 - a ticket dispenser
 - a painting application



Discussion

- The following are criteria for the usability goals of a ticket dispenser:
 - Learnability: can a user learn to use the system for the first time in **less than 2-3 minutes**?
 - Memorability: if a user of the ticket dispenser has not used the dispenser in **2 months**, will he/she be able to user it properly again?
 - Efficiency: will the user of the ticket dispenser be able to select, purchase and get a ticket **in less than a couple of minutes**?
- User experience goals:
 - Motivating, aesthetically pleasing, enjoyable

Discussion

- The following are criteria for the usability goals of a paint application:
 - Learnability, efficient, memorable , utility
 - Support creativity, aesthetically pleasing

Discussion

- Think about one appliance/device at your home.
 - What would be two important usability goals for the device?
 - Translate the core goals outlined above into two criteria (specific and measurable)
 - Do you feel the designers of these appliances satisfied the goals you outlined above?

Question

- Provide an example of applications in which several user experience goals were 'sacrificed' for the benefit of the overall usability.
- And vice versa, think of a couple of applications in which the usability goals may have not been met for the purpose of enriching the user's experience.

Answer

- An example of the first: online e-transfer is not **fun** and **enjoyable** but yet more **efficient** than driving the car to the bank, finding a spot to park, waiting in line and making the transfer.
- An example of the second: a drawing utility used by artists, which can support **creativity** (user experience goal) but will not provide the same level of **efficiency** or even as **effective** as done manually.

Generic Design Principles

- **Generic “rules of thumb”** that describe features of “usable” systems
- Guidelines of **what to include** and **what not to include** when designing an interface
- Derived from practice, theory, and research
- Used later in heuristic evaluation

Design Principles

- Design principles are used by interaction designers to aid their thinking when designing for the user experience
- More specifically, they are intended to help designers explain and improve their designs
- A number of design principles have been promoted
- We will briefly cover the most common ones

Generic Design Principles

- Visibility
- Feedback
- Constraints
- Mapping
- Consistency
- Affordance
- Simplicity
- Matching
- Minimizing memory load
- Diagnose/Recover errors
- Control and freedom
- Flexibility
- Provide Help

Design Principle - Visibility

- Visibility
 - Visibility is the basic principle that the more visible an element is, the more likely users will know about them and how to use them
 - Advocates the use of making core user functions clearly apparent, hides secondary user functions
 - Visible properties guides users into what to do next

Design Principle - Visibility

- Problems arise when we cannot “see” how to do use a device
- Sensor technology like auto faucets - not sure how to use - guess where to put hands
- Visible knobs, dials and buttons have been replaced by invisible and ambiguous “active zones”



Windows 8



where is the start button in windows 8

All Videos Images News Maps More

About 551,000,000 results (0.76 seconds)

First, in **Windows 8.1**, the **Start button (Windows button)** is back. It's there in the lower-left corner of the desktop, right where it always was. (It even appears in TileWorld if you point your mouse to that corner.) The **Start button** does not open the traditional **Start menu**, however. Apr 13, 2014



[How to Get the Real Start Menu Back in Windows 8 or 8.1 - Ya](https://www.yahoo.com/.../how-to-get-the-real-start-menu-back-in-windows-8-or-)
<https://www.yahoo.com/.../how-to-get-the-real-start-menu-back-in-windows-8-or->

About this result

HOME > COMPUTING > HOW TO BRING BACK THE START MENU AND BUTTON TO WINDOWS 8

How to bring back the Start menu and button to Windows 8

By Jamie Lendino on March 24, 2015 at 9:50 am | 456 Comments

f t G+ Y 3.6K SHARES



With Windows 10 seemingly just around the corner, Microsoft plans to fix one of its most egregious mistakes with Windows 8: the missing Start menu. By now you may know the story, as the company has gradually backtracked throughout various updates since the OS launched in late 2012. And **Windows 10** will bring the real thing back, albeit

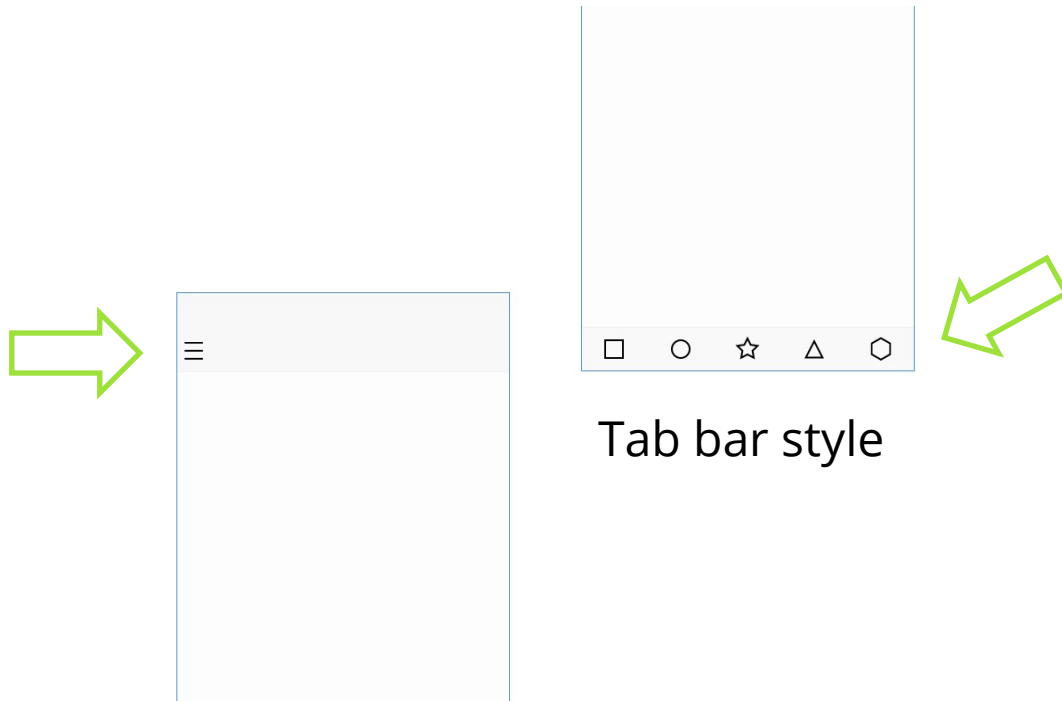
in an enhanced fashion that doesn't depend on fiddly per-user shortcuts and extra folders.

The thing is, unless you're willing to **experiment with Windows 10 Technical Preview**, the new version won't help you yet. For several years now, every Windows desktop and laptop sold in a store comes with Windows 8. While Windows 8.1 Update somewhat improves the situation, it's still no Windows 7 when it comes to ease of use.

Here's the problem, in a nutshell: Let's say you've just installed Windows 8.1 for the first

Design Principle - Visibility

- From hamburger menus to tab-bar menus to improve the visibility of their key experiences

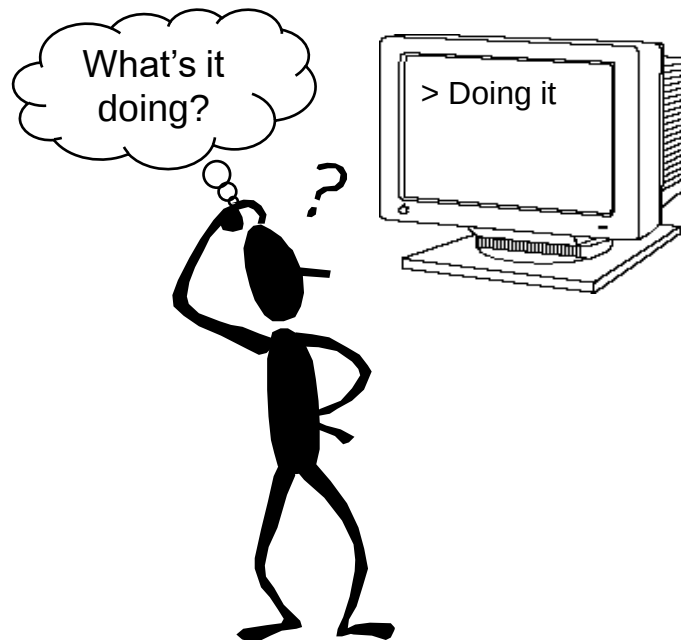


Hamburger



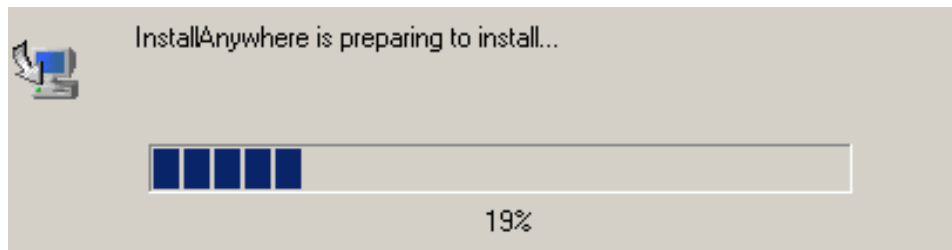
Design Principle - Feedback

- Feedback
 - continuously inform the user about the system's state
 - how it is interpreting the user's input
 - user should at all times be aware of what is going on

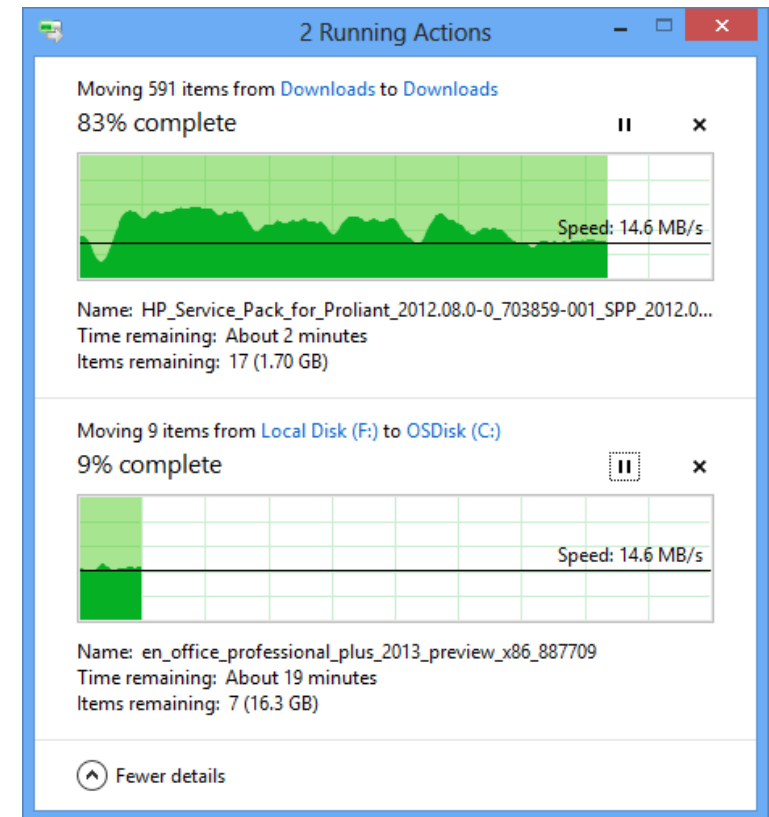


Design Principle - Feedback

- Various kinds of feedback are available for interaction design
 - Audio (Verbal, non-verbal)
 - Tactile
 - Visual
 - And combinations of these



No indication of the files being installed but still gives %



Multiple files being copied, feedback is file by file.

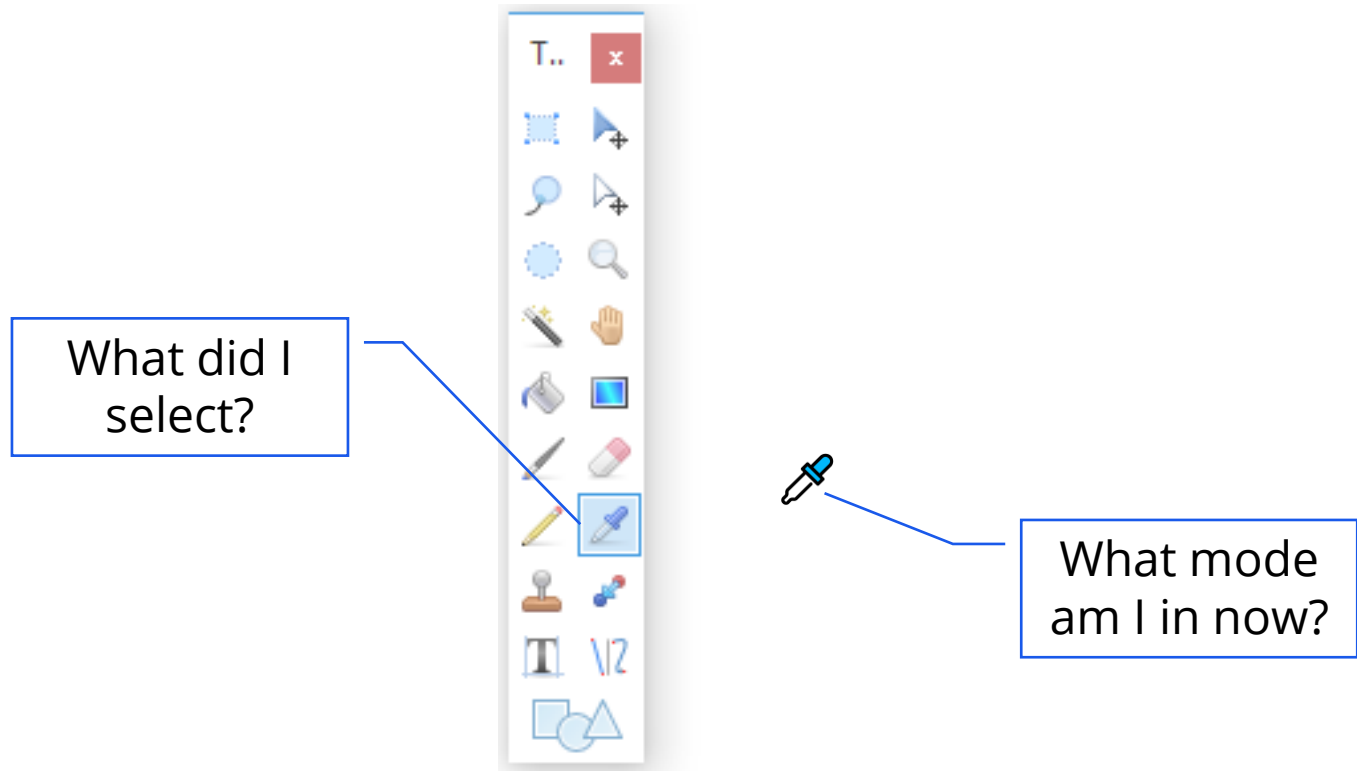
Design Principle - Feedback



Reduce users' perception of time

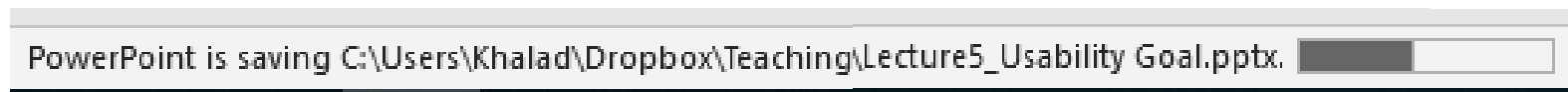
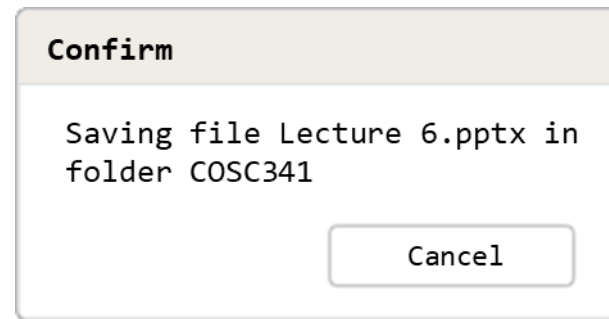
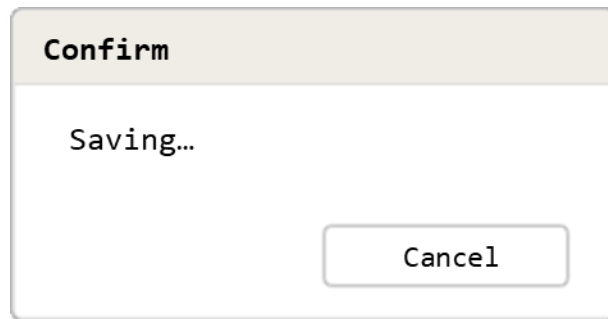
Design Principle - Feedback

- Paint.net



Design Principle - Feedback

- Should be as specific as possible, based on user's input
- Best within the context of the action

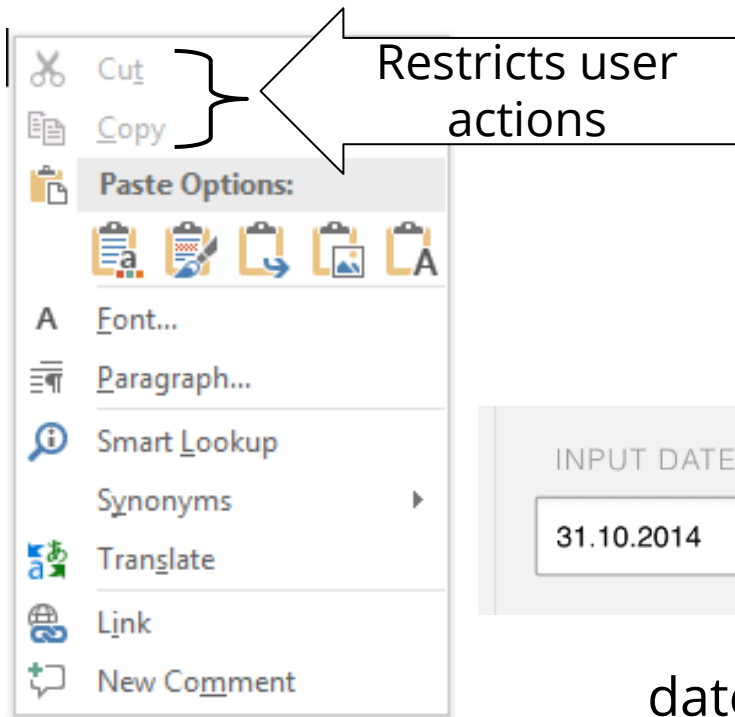


Design Principle - Feedback

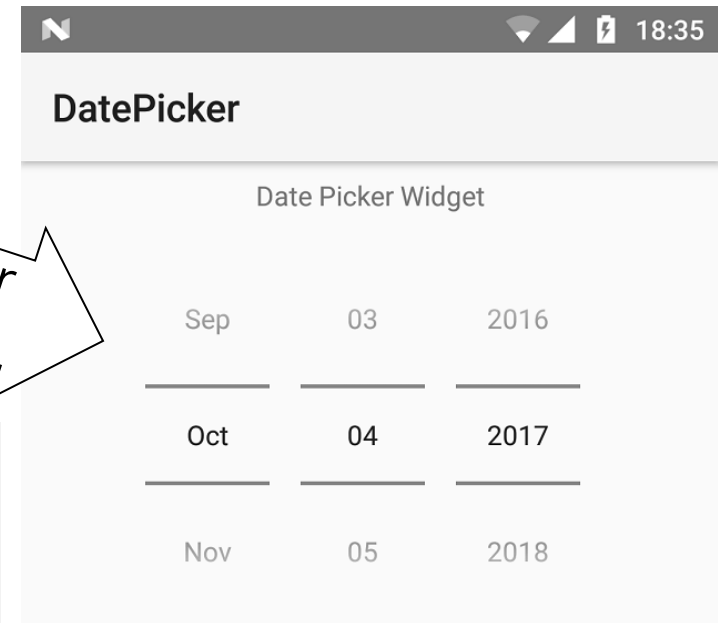
- Response time
 - how users perceive delays
 - 0.1 second max: perceived as 'instantaneous'
 - 0.7 second max: visual-motor reaction time to unexpected events
 - 10 seconds: limit for keeping user's attention focused on the dialog
 - > 10 seconds: user will want to perform other tasks while waiting, i.e. get a cup of coffee

Design Principle - Constraints

- Constraints
 - restricts the kinds of user actions that can take place for any given mode of interaction
 - A common design practice in graphical user interfaces is to deactivate certain menu options by shading them gray



date picker



date picker

Design Principle - Constraints

The diagram illustrates design constraints in a form layout. It shows two versions of a form side-by-side, separated by a vertical line. The left version shows a form with labels and input fields, while the right version shows the same form with the labels and input fields aligned to the right, creating a consistent visual structure. A large arrow labeled 'Constraints' points from the right version back to the left version, indicating that the constraints are applied to the original form design.

Left Form (Original):

- Enable D
- Name :
Ex: Albert
- Email :
Ex: albert@gmail
- Message :
Ex: Hi! Wass

Right Form (Constrained):

- Name :
James
- Email :
James@rediffmail.com
- Message :
Hello!!!!
- Submit

Bottom Form (Constrained):

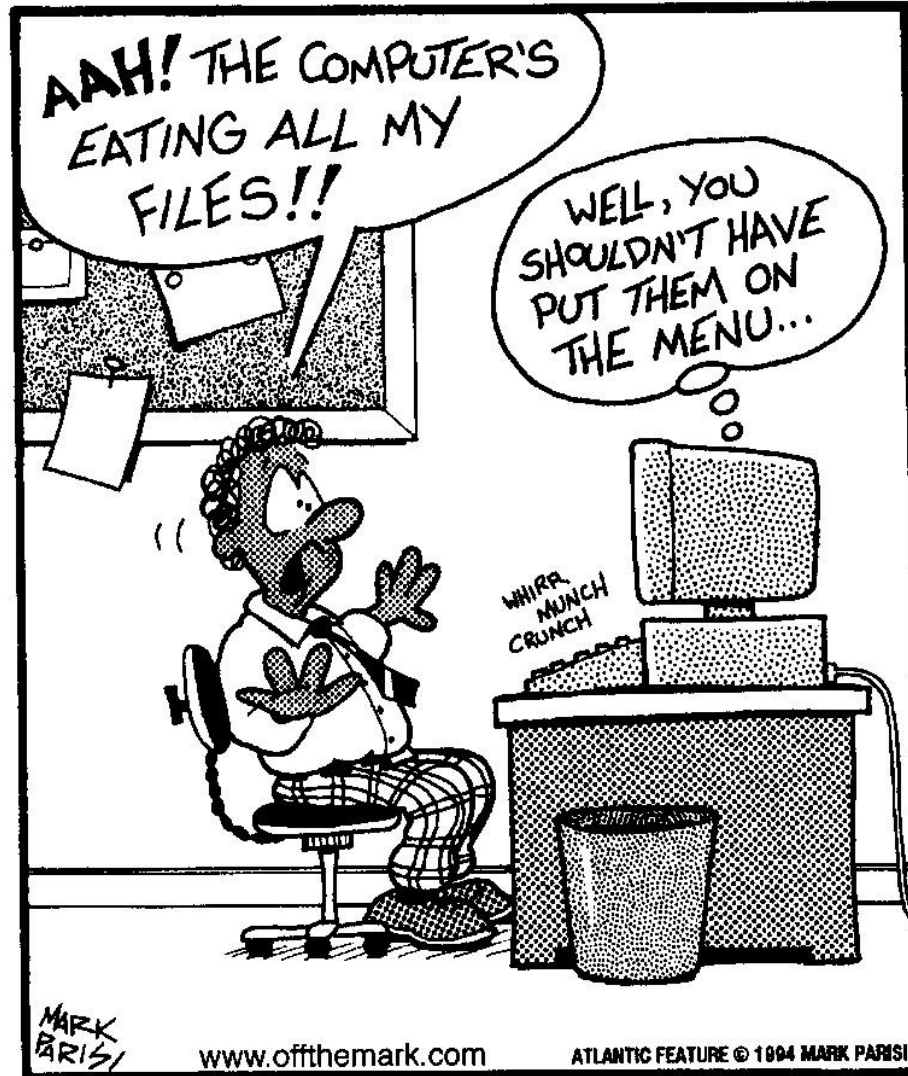
- Submit

Constraints:

Design Principle - Constraints

off the mark

by Mark Parisi

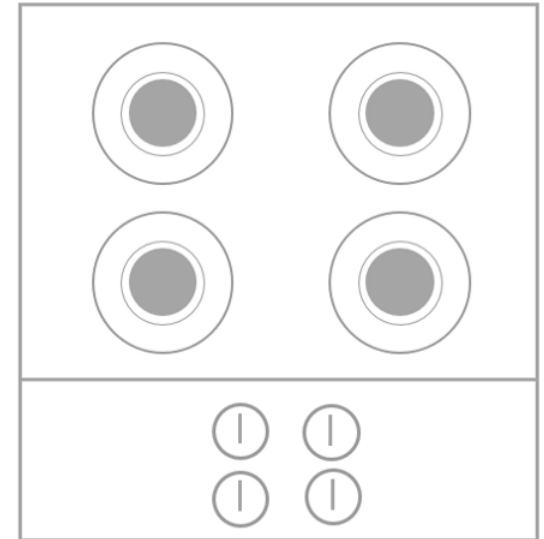
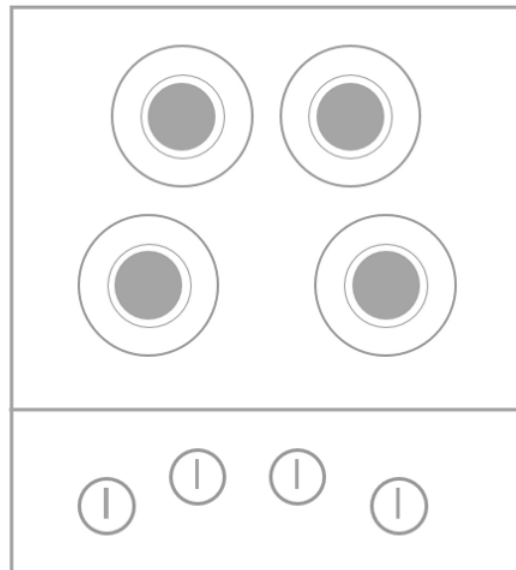
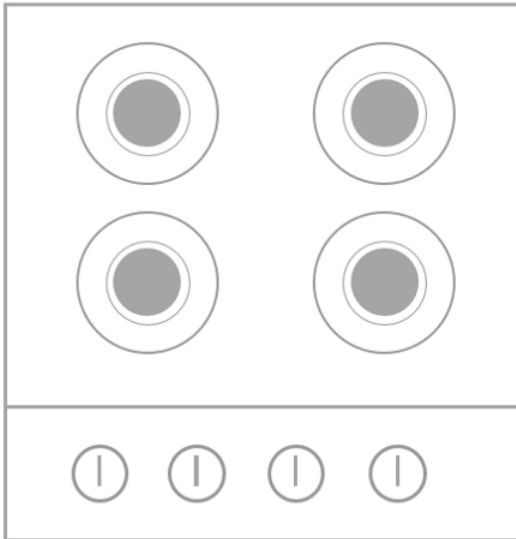


Design Principle - Mapping

- Mapping
 - relate controls to the intuitive understanding of how they should be used
 - e.g., mapping of down and up arrows on a scroll bar to the movement of a document on a screen
 - relationship to controls and their effect



Design Principle - Mapping



Design Principle - Consistency

- Designing interfaces to have similar operations and use similar elements for achieving similar task
- Systems are usable and learnable when similar concepts are expresses in similar ways
- Enables people to quickly transfer prior knowledge to new contexts and focus on relevant tasks



Design Principle - Consistency

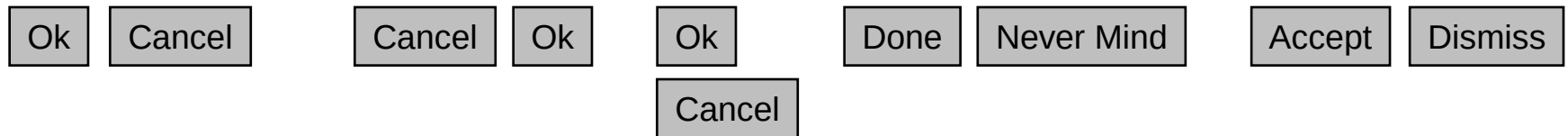
- Consistency

- Consistent results

same words, commands, actions will always have the same effect in equivalent situations - leads to predictability

- Consistent language and graphics

same information/controls in same location on all screens / dialog boxes



- Consistent input

consistent syntax across complete system

"Ctrl+C"/"Ctrl+V" for Copy/Paste in Windows

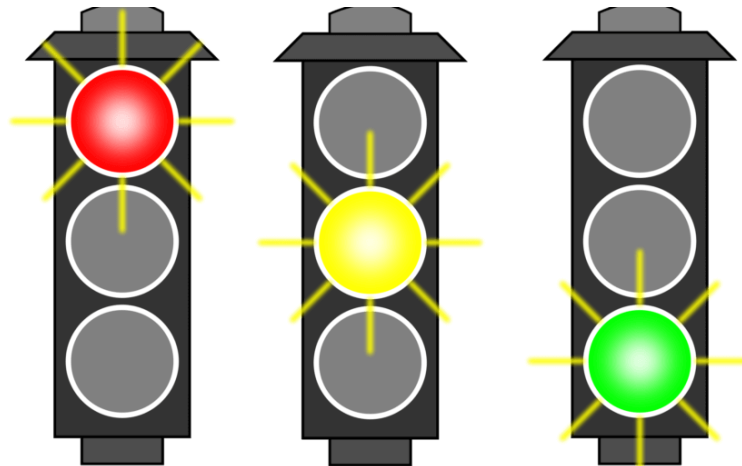
Design Principle - Consistency

- Aesthetic
 - Style and appearance is repeated to enhance recognition, communicates membership and sets emotional tone
 - Mercedes Benz vehicles are instantly recognizable because the company consistently feature its logo on all its vehicles
 - Associated with quality and prestige; respected



Design Principle - Consistency

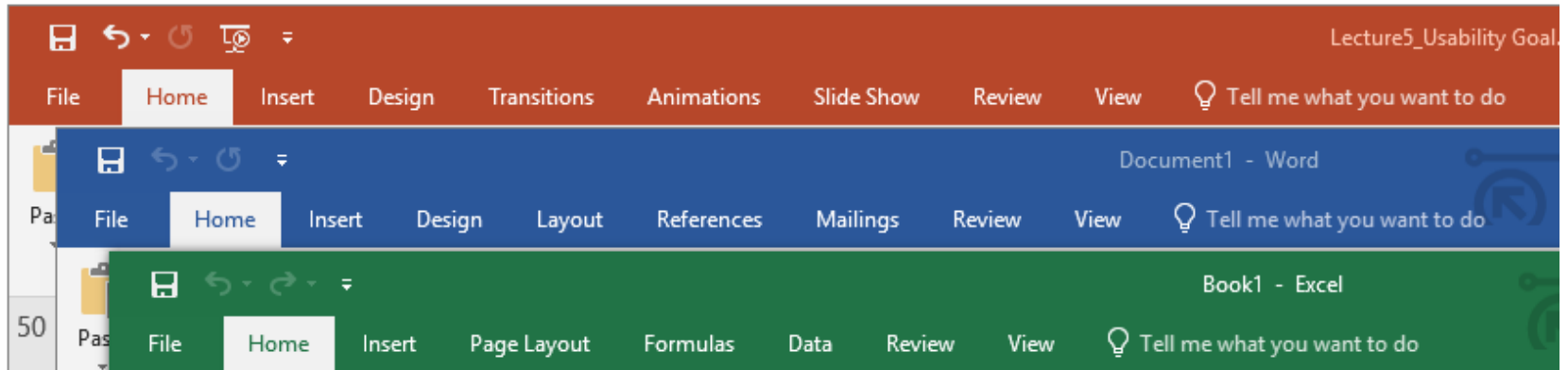
- Functional
 - Meaning and action are consistent to improve learnability and understanding
 - Consistent use of symbols to represent similar concepts, leverages prior knowledge and makes new things easier to use
 - Traffic always turns yellow before red



Design Principle - Consistency

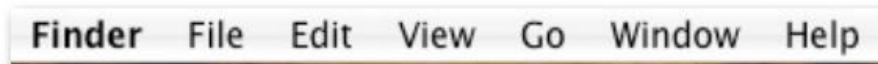
- Internal

- Is the consistency of elements within the same system (e.g. road signs in Canada are consistent across different parts of the country).
- Internal consistency helps cultivate trust and communicates that a system is designed as a whole rather than a random collection of pieces



Design Principle - Consistency

- External
 - consistent with other elements in the environment
 - extends the benefit of internal consistencies across multiple, independent systems
 - more difficult to achieve because different systems rarely observe the same design standards



Finder File Edit View Go Window Help



Safari File Edit View History Bookmarks Window Help



Photoshop File Edit Image Layer Select Filter View Window Help

Design Principle - Affordance

- Affordance
 - At a very simple level, to afford means "to give a clue"
 - Perceived and actual properties of an object that give clues to its operation



Design Principle - Affordance

- Appearance indicates how the object should be used
 - chair for sitting
 - table for placing things on
 - knobs for turning
 - slots for inserting things into
 - buttons for pushing



Perceived Affordances



Design Principle - Affordance

What is your favorite color?

- ☒ Red
- ☐ Green
- ☐ Blue
- ☐ Other

One item can be selected

What is your favorite pet?

- ☒ Cats
- ☒ Dogs
- ☐ Birds

Multiple items can be selected

Example of a Bad Affordance

- Handles are for lifting, but these are for scrolling



- from AudioRack 32, a multimedia application

Design Principle - Simplicity

- Common tasks should be easy to perform
- Provide only the necessary, “less is more”
- Don’t give users extra problems to solve
- Remove or hide irrelevant or rarely needed information
 - competes with important information on screen

Design Principle - Simplicity

The screenshot shows the old Godaddy website, which is highly cluttered and lacks simplicity. The layout is divided into several sections:

- Header:** Contains the Godaddy logo, a "Reseller Plans! RELOADED" banner, and a "Verify the site for your protection!" link. Navigation links include "My Account", "Support", "My Cart", and "Check Email".
- Navigation Bar:** A long bar with links for "Domain Names", "Hosting & Servers", "Site Builders", "Email", "SSL Certificates", "Ecommerce", and "Be a Reseller". Below this is a secondary bar with links like "My Account / Logout", "Company Info", "Why our prices are so low", "10-Point Value Plan", "What's New!", "FAQ", "Link To Us", "XML", "NEW! RSS Feeds".
- Left Sidebar:** A vertical menu with links for "Domain Names" (Register new, Bulk register, Transfer, Make Private, Backorder, NEW! Sell your domain name, NEW! View domain auctions), "Web Hosting" (Up to \$47 off, See All Hosting Plans, Virtual Dedicated Servers, Dedicated Servers), "Site Builders" (Save 20%, WebSite Tonight! (Online), WebSite Complete!), "Email Accounts" (See All Email Plans, Online File Folder, Fax Thru Email, Online Group Calendar), "Secure SSL Certificates" (Turbo - Only \$29.95!, High Assurance, Wildcard), "Ecommerce" (Merchant Accounts, Quick Shopping Cart™, Traffic Blazer®, Express Email Marketing™, News-Blazer™, c-Site®, Reseller Opportunities), and "Go Daddy T-Shirts".
- Main Content Area:** Features a large "Secure Login" form, a "Radio GoDaddy" section, and a prominent "Secure email accounts start at just \$9.95/yr!" banner. Below this is a "Find a domain name now:" search bar with a ".com" dropdown and a "Search" button. A list of domain extensions (.com, .us, .biz, .info, .net, .org, .ws, .name, .tv, .co, .de, .jp, .be, .at, .uk, .nz, .cn, .tw) is provided.
- Bottom Section:** Contains three columns of promotional offers: "Web Hosting & Servers" (Up to 4GB space & 200GB transfer from \$3.95/mo!), "Spam, Virus & Fraud Protected Email" (NEW! Blockbuster plans from just \$9.95/yr!), and "Traffic Blazer™" (Improve site traffic and sales from \$29.95/yr!). Each column includes a "Click to learn more" link.
- Footer:** A green bar with "View Our Catalog" and "See More Hot Deals & Special Offers!". Below this is a contact bar with "24/7 Sales and Support: (480) 505-8877", "Billing Questions? Call (480) 505-8855", and "Free e-News: Enter email address" with a "GO" button.

Design Principle - Simplicity

GoDaddy

Domains

Websites + Marketing

Templates

Contact Us 24/7

Blog

Help

Sign In

Find your perfect domain

Website Builder

Create your site.

Start for Free

No credit card required.



Domain Names

.ca for just C\$0.01/1st yr.

Find Your .ca

2-year purchase required. Additional year(s) C\$19.99

.com C\$2.99*/1st yr	Web Hosting C\$5.99/mo	SSL Security C\$84.99/yr	Email C\$3.99/mo	 Preferred by 83% of Canadian shoppers. C\$0.01/1st yr.
WordPress C\$4.99/mo	Office 365 C\$9.99/mo	Pro Web Design	SEO C\$9.99/mo	



Canada's unhappiest cities revealed

Those in rural areas are quite content, but two major metropolises are failing - despite being among the world's most livable.
'Sense of belonging' is key »

245 people reacting



Buffalo Bills star's halftime retirement will cost him...



'I had accepted that I was going to die'



'She didn't want to die without us getting married'



Arrest made in 'horrific' cold-case rapes



Passing out résumés at intersection pays off



Entertainment Entertainment Tonight

Prince Harry Proves He's Still the Cheekiest Royal When Caught Sneaking Food at Meghan Markle's Cookbook Event

The 34-year-old royal was caught red-handed.



Sponsored Casper

Why The Casper Mattress is Selling Out in Canada

It ships for free and you can try it at home for 100 nights. Did we mention there's also a napmobile that traveled from Toronto to Calgary?



Lifestyle Marie Claire

Kate Middleton's Net Worth Before She Married Prince William Will Make Your Jaw Drop

Hums "Bills, Bills, Bills"

Trending Now

1. Rod Rosenstein
2. Philippe Couillard
3. Tornado warnings
4. Doug Ford
5. Donald Trump
6. Millie Bobby Brown
7. Jeff Banister
8. Stephen Colbert
9. Toronto Raptors
10. Netflix's Maniac

Advertisement

Protect your business & data with next-generation firewall

Only \$1436.99*

PCM Canada | CISCO

*Subject to change

Kelowna, BC

Today



14° 8°

Sun



14° 8°

Mon



17° 7°

Tue



18° 7°

Featured Picks



Woman fell asleep on her commute and was locked on a GO Train

Yahoo Canada Original Videos



VOTE: Do you think 'Sesame Street' missed an opportunity with Bert and Ernie?

Yahoo Canada News

[Gmail](#) [Images](#)



[Sign in](#)



Google Search

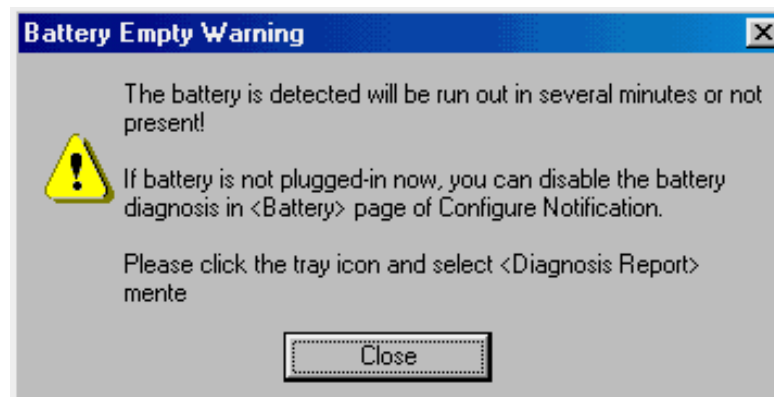
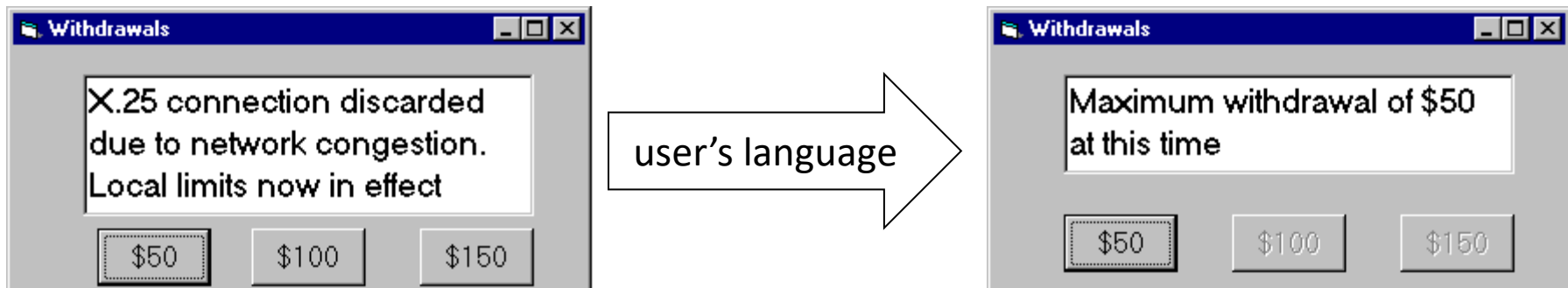
I'm Feeling Lucky

The world is closer than ever. [Google Translate](#), now in over 100 languages.

Google.ca offered in: [Français](#)

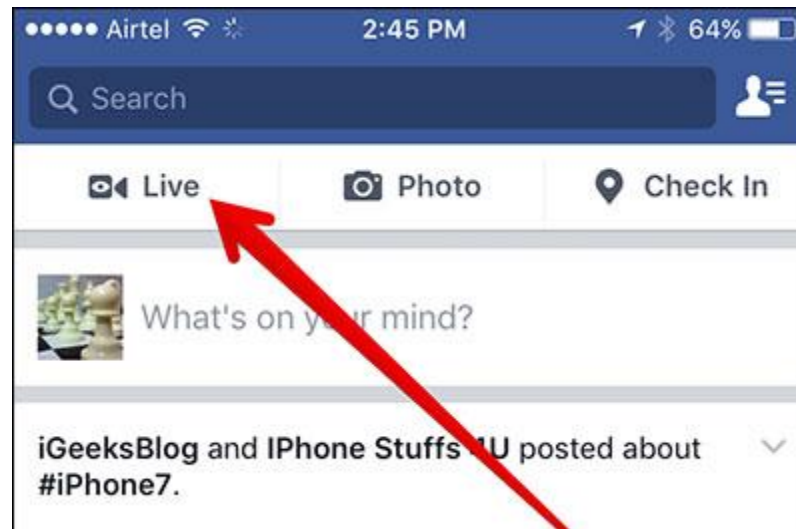
Design Principle - Matching

- Match between system and real world
 - speak the user's language
- Terminology based on users' language for task
- e.g. withdrawing money from a bank machine



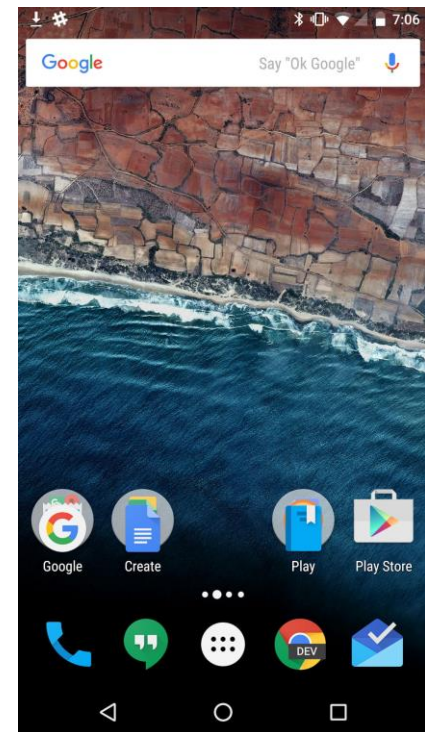
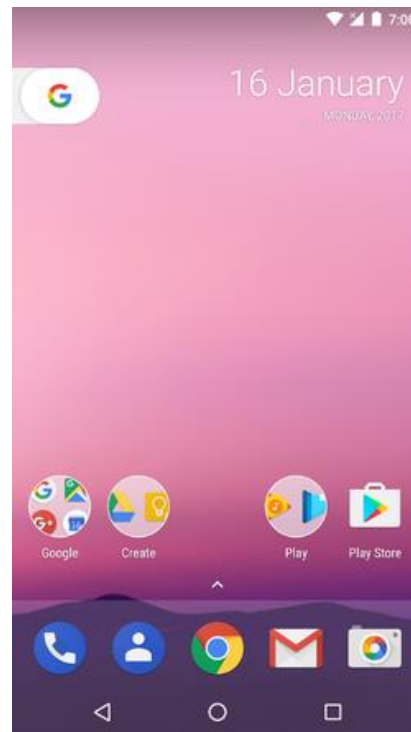
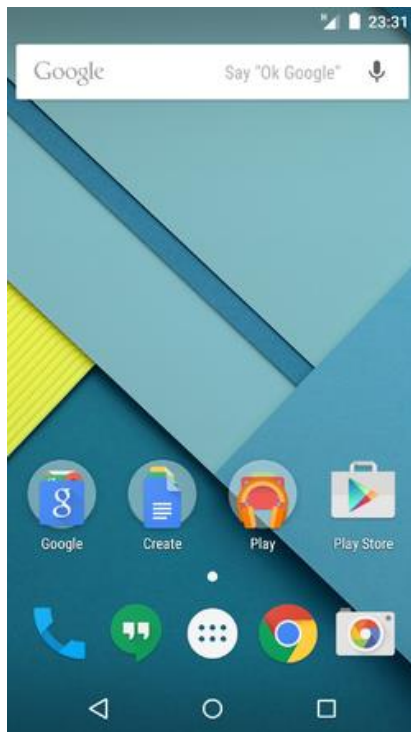
Design Principle - Matching

- Use meaningful mnemonics, icons, and abbreviations



Design Principle - Minimizing memory load

- Build on existing mental models
- People already have mental models about how a system work, based on their past experiences of using the system
- Promote recognition over recall



Design Principle – Diagnose/Recover errors

- Errors are inevitable, users will make them!
- Errors we make
 - Mistakes:
arise from **conscious deliberations** that lead to an error instead of the correct solution
 - Slips:
unconscious behavior that gets misdirected en route to satisfying a goal, e.g. drive to store, end up in the office

Slip vs. Mistake



- Imagine you are using a mapping application wanted to find something, and clicked this icon.
- Let's pretend that in the application, the icon meant to magnify.
- Is this a slip or a mistake?

General Guidelines

- Include helpful constraints

The screenshot shows a flight booking interface with a date selection calendar. The calendar is titled "Select depart date" and displays two months: July 2015 and August 2015. The July calendar has a grid with days of the week (S, M, T, W, T, F, S) and dates. The date 8 is highlighted in blue, and the date 11 is highlighted in green. A hand is shown pointing at the date 15, which is circled. The August calendar also shows a grid with days of the week and dates. Below the calendar, there are input fields for "BOS" (Boston Logan, MA - BOS) and "LGA" (New York (LaGuardia), NY - LGA). The departure date is set to "07/08" (Depart Wed, Jul 8 2015) and the return date is set to "07/11" (Return Sat, Jul 11 2015). There are also input fields for the number of adults (1) and seniors (0). At the bottom, there are links for "Where we fly", "Map search", and "Flexible dates", a "Promo code" field, an "Advanced Search" link, and a "Search" button.

Select depart date

July 2015

S	M	T	W	T	F	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

August 2015

S	M	T	W	T	F	S
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31					

Book

☒ Round trip

BOS **LGA** **07/08** **07/11** **1** **0**

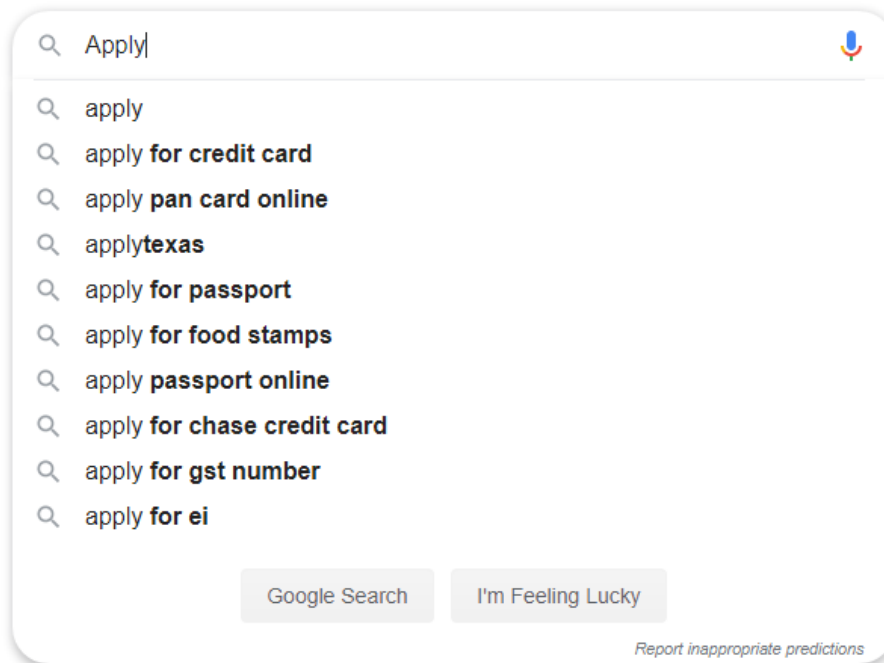
Boston Logan, MA - BOS New York (LaGuardia), NY - LGA Depart Wed, Jul 8 2015 Return Sat, Jul 11 2015 Adults Seniors

Where we fly Map search Flexible dates

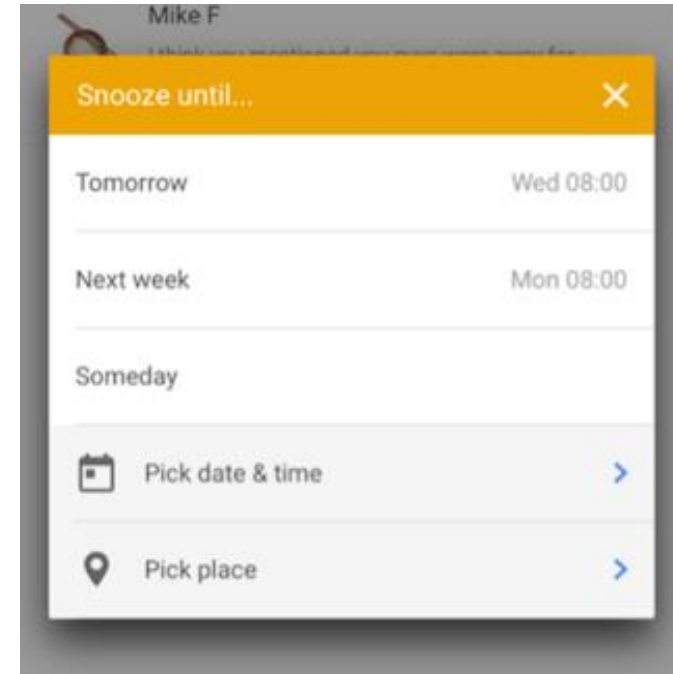
Promo code [Advanced Search](#) **Search**

General Guidelines

- Offer suggestions



- Choose good defaults



General Guidelines

- Use forgiving formatting

Profile

* NAME

First Name Last Name

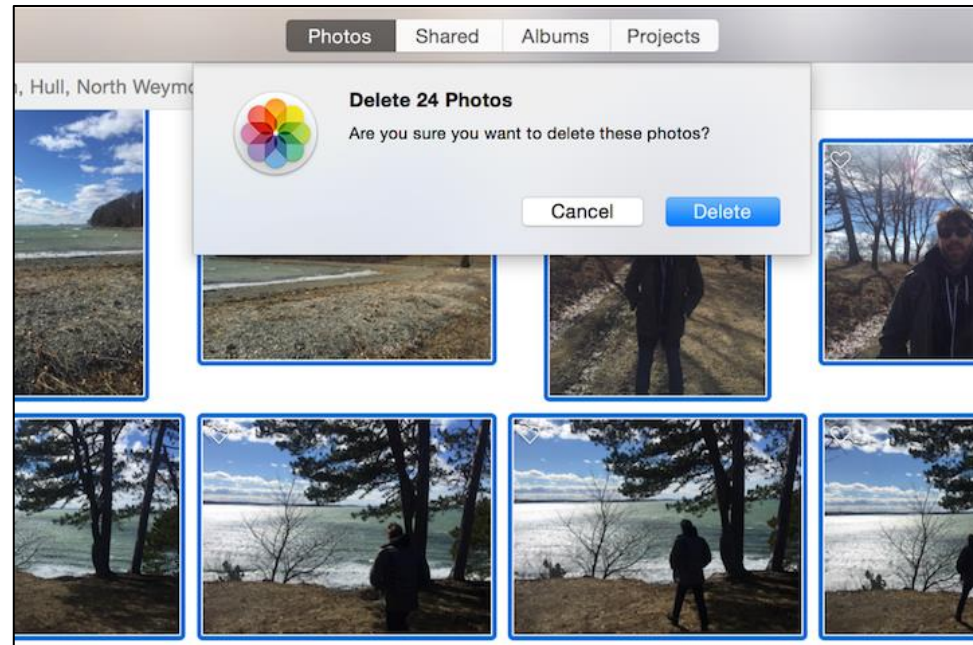
* MOBILE NUMBER

+1 (555) 666-7778

* LANGUAGE

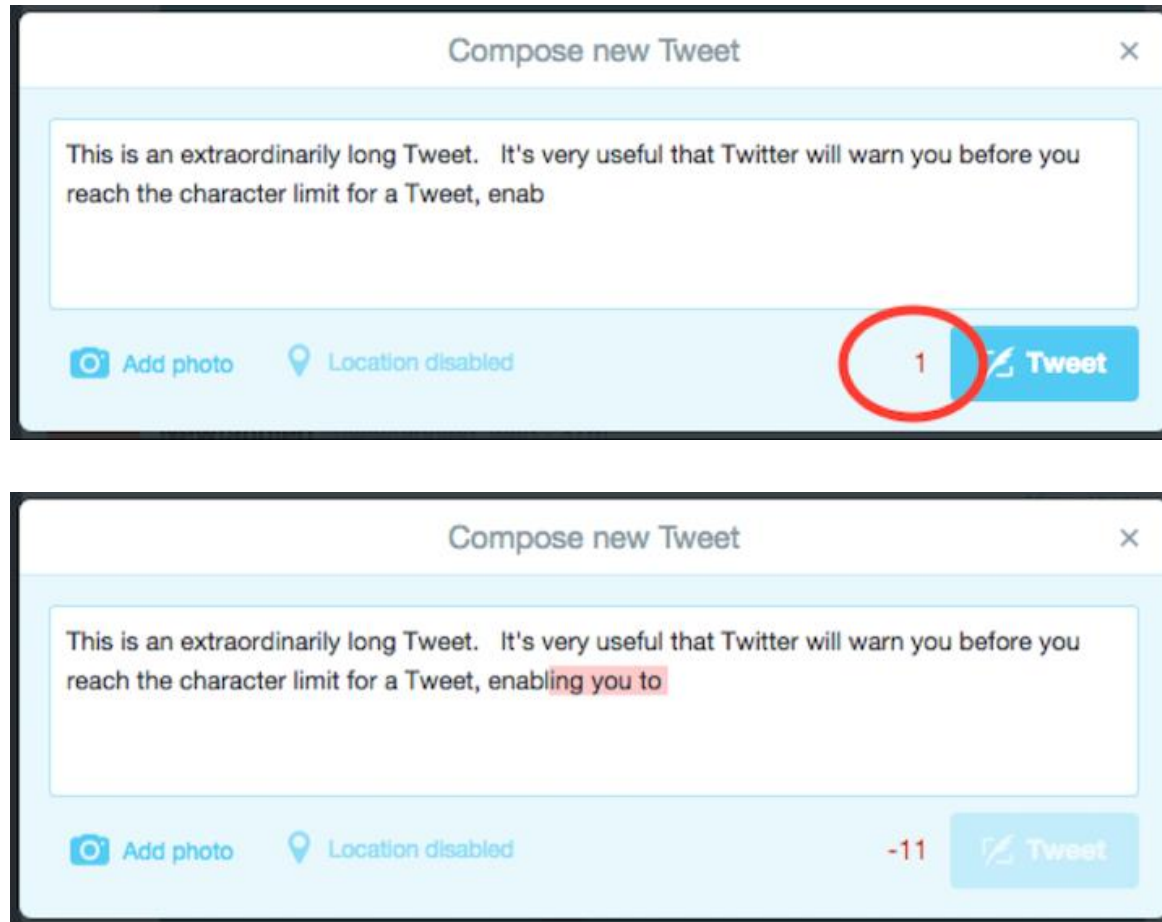
English (United States)

Confirm before destructive actions



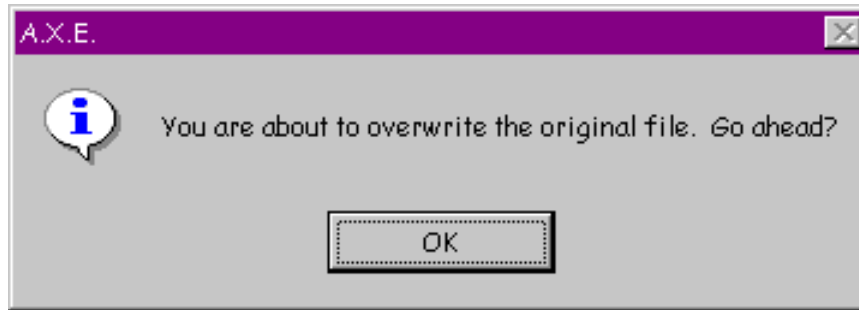
General Guidelines

- Warn before errors are made



- Example: twitter character count warning

Provide Meaningful Error Messages



No other choices provided!

"This action will not save the game"



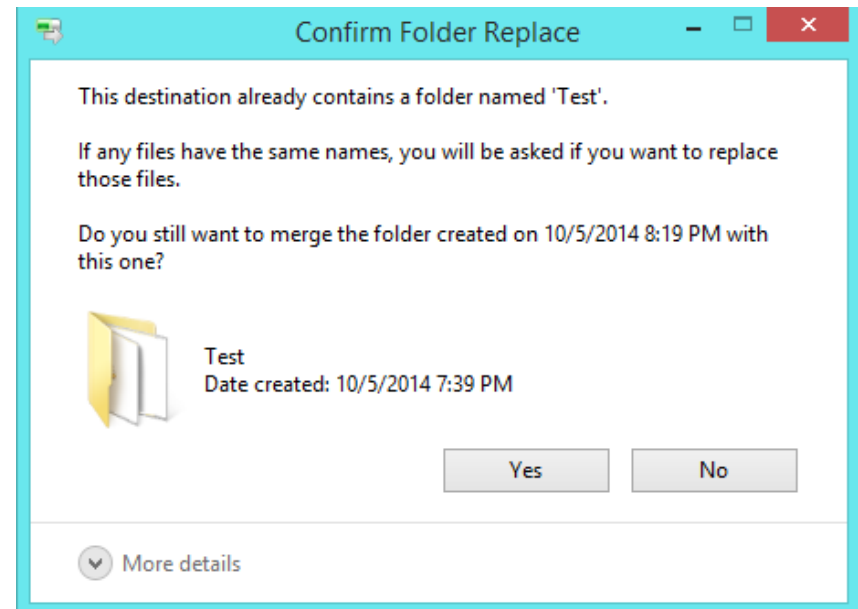
Provide Meaningful Error Messages

We have removed your email address from our list.

We're sorry to see you go.

Was this a mistake? Did you forward one of our emails to a friend, and they clicked the unsubscribe link not realizing they were in fact unsubscribing you from this list? If this was a mistake, you can re-subscribe at:

Subscribe



Guidelines for Error Messages

- Shneiderman's guidelines for error messages include:
 - Have a positive tone: avoid using terms like **FATAL, INVALID, BAD**
 - Be specific and address the problem in the user's terms: avoid obscure internal codes (Error code 102E: ...)
 - Put users in control: tell them what they should do to recover and continue
 - Use a consistent interface and comprehensible format
 - Provide context-sensitive help

Discussion

- Re-write the following common error messages using a friendlier language. The message should explain the cause and suggest a method for fixing the problem. For each message imagine a context in which the problem will occur:
 - SYNTAX ERROR
 - INVALID FILE NAME
 - INVALID DATA
 - DRIVE ERROR: ABORT, RETRY OR FAIL?

Activity - Answer

- There is a problem with the way you have written the command. Check for typos.
- You cannot use '/' or '?' in a file name. Please choose another file name.
- This field will only accept numeric data. Try again, checking that only numbers are used.
- There is a problem with reading your disk. Try inserting it again.

SYNTAX ERROR

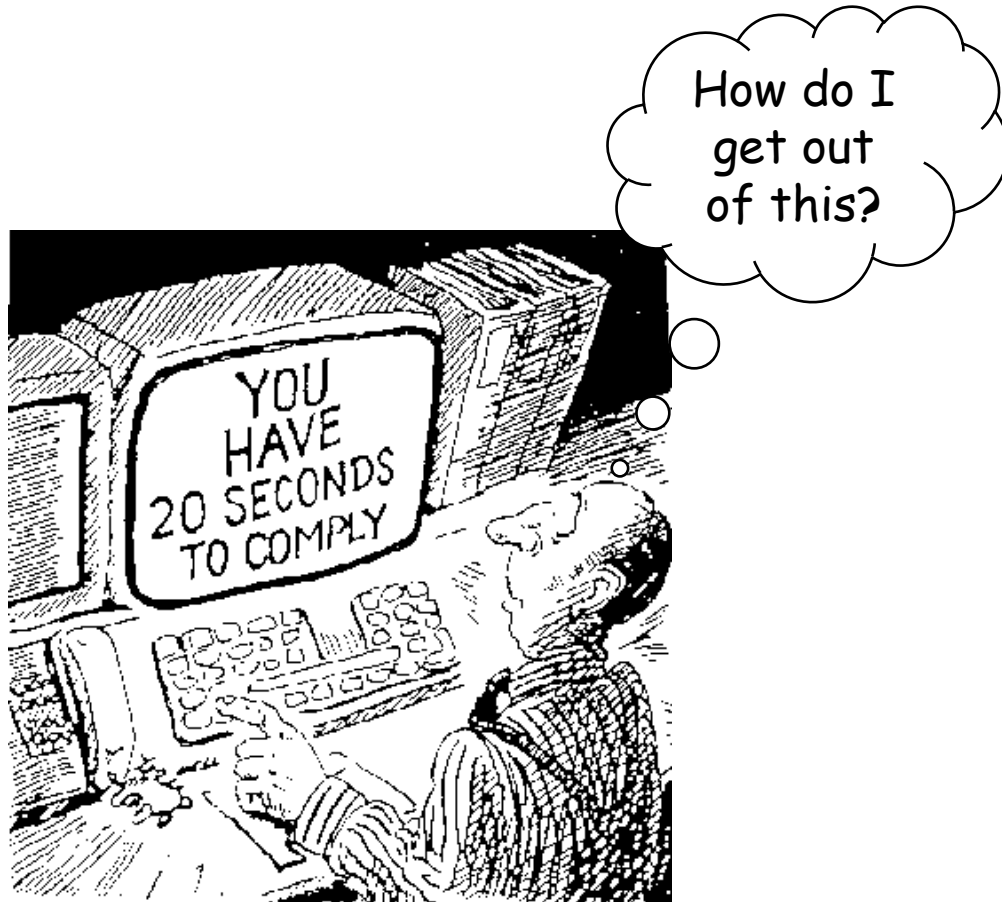
INVALID FILE NAME

INVALID DATA

DRIVE ERROR: ABORT, RETRY OR FAIL?

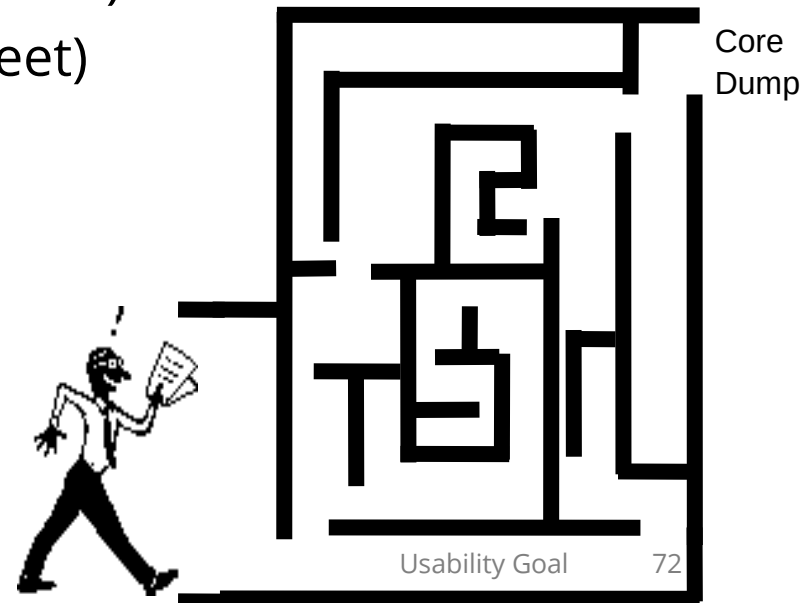
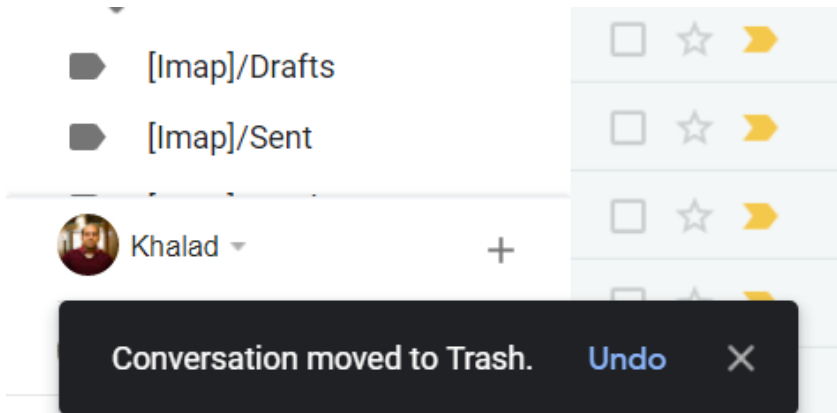
Design Principle - Control and freedom

- Use a clearly marked "emergency exit" to leave the unwanted state without having to go through an extended dialogue



Design Principle - Control and freedom

- Users don't like to feel trapped by the computer!
 - should offer an easy way out of as many situations as possible
- Strategies:
 - **Cancel button** (for dialogs waiting for user input)
 - **Undo** (can get back to previous state)
 - **Interrupt** (especially for lengthy operations)
 - **Quit** (for leaving the program at any time)
 - **Defaults** (for restoring a property sheet)



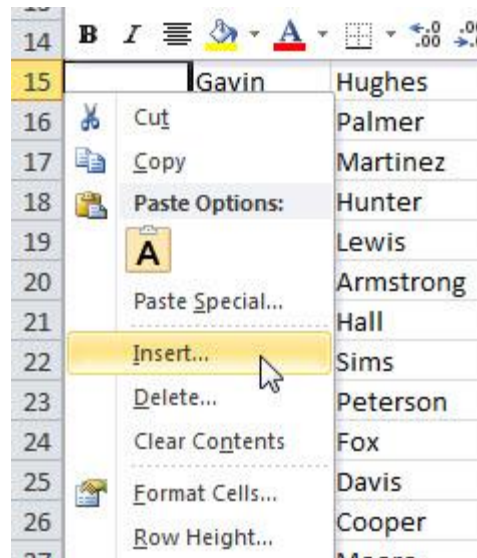
Design Principle - Flexibility

- The design accommodates a wide range of individual preferences and abilities.
- Example: a museum that allows visitors to choose to read or listen to the description of the contents of a display case.



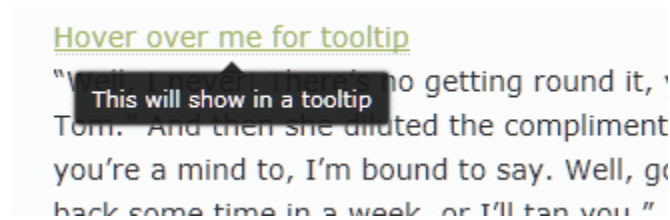
Design Principle - Flexibility

- Shortcuts: Experienced users should be able to perform frequently used operations quickly
- Strategies:
 - Provide keyboard and mouse accelerators (commands, function key, double click)
 - Support type-ahead (entering input before the system is ready for it)

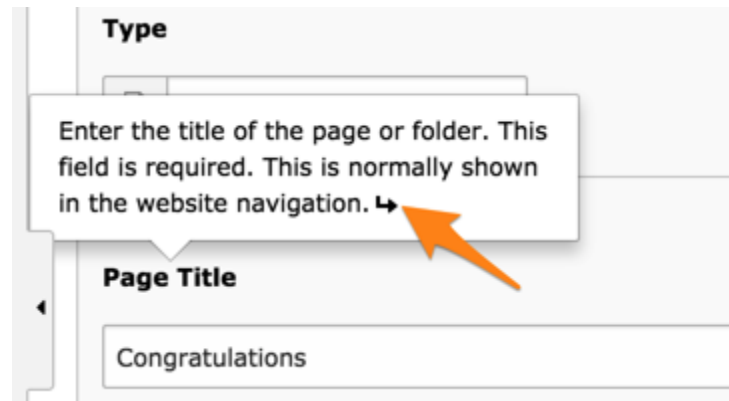


Design Principle - Provide Help

- Tutorial and/or getting started manuals
- Reference manuals
- Reminders
- tooltips (hints)

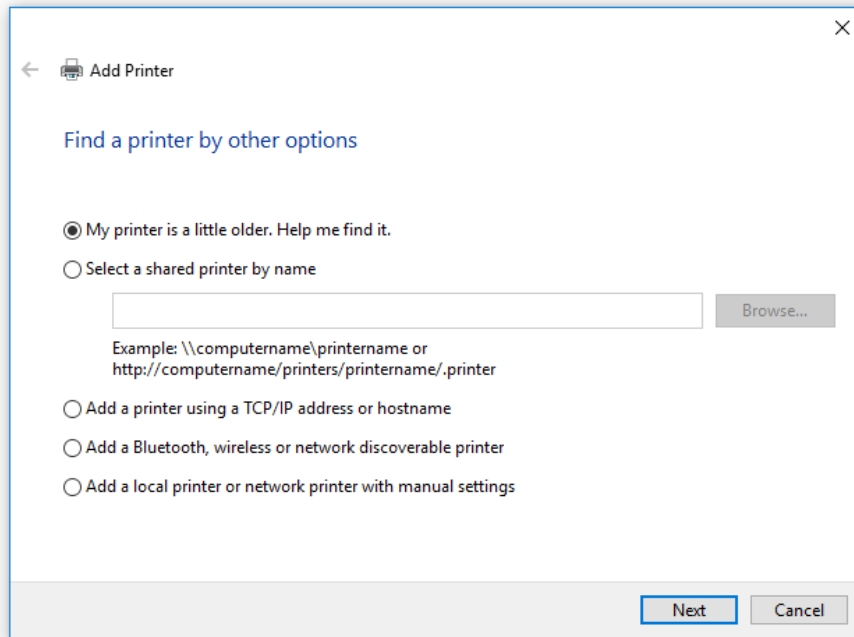


- Context-sensitive help



Types of help

- Wizards
 - Walks user through typical tasks



← Add Printer

Find a printer by other options

☒ My printer is a little older. Help me find it.

☐ Select a shared printer by name

Browse...

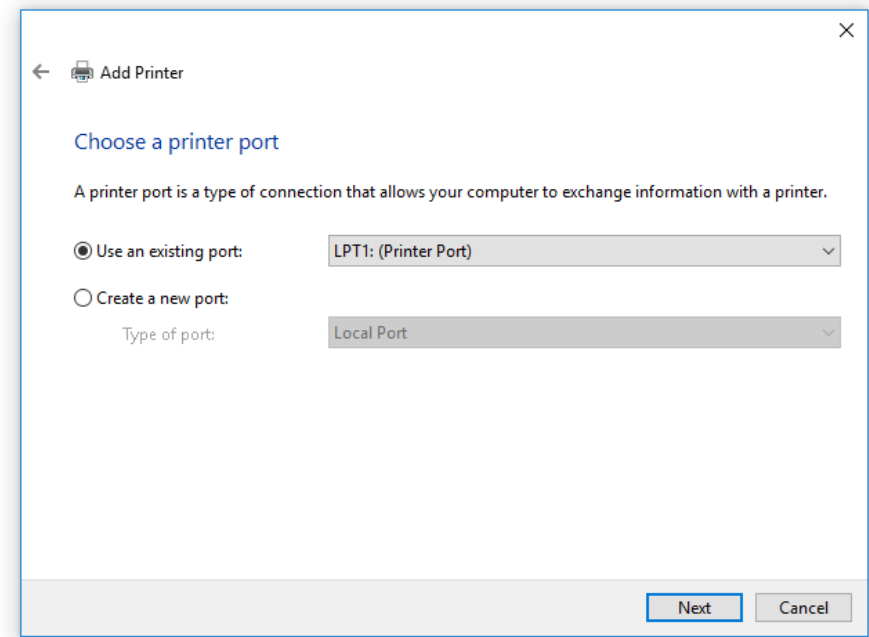
Example: \\computename\printername or
http://computename/printers/printername/.printer

☐ Add a printer using a TCP/IP address or hostname

☐ Add a Bluetooth, wireless or network discoverable printer

☐ Add a local printer or network printer with manual settings

Next Cancel



← Add Printer

Choose a printer port

A printer port is a type of connection that allows your computer to exchange information with a printer.

☒ Use an existing port: LPT1: (Printer Port) ▼

☐ Create a new port:

Type of port: Local Port ▼

Next Cancel

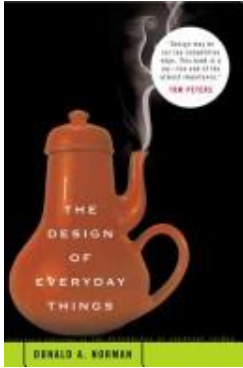
From Principles to Goals

Usability Principle	Usability Goal(s)
Visibility	Efficiency
Feedback	Memorability
Constraints	Efficiency, safety
Mapping	Effectiveness, efficiency, learnability, memorability
Consistency	Learnability, efficiency, memorability
Affordance	Learnability
Simplicity	Memorability, efficiency, learnability, good utility
Matching	Efficiency
Minimizing memory load	Memorability, learnability, efficiency
Diagnose/Recover errors	Safety
Control and freedom	Safety, memorability
Flexibility	Efficiency, memorability
Provide Help	Safety, efficiency, learnability

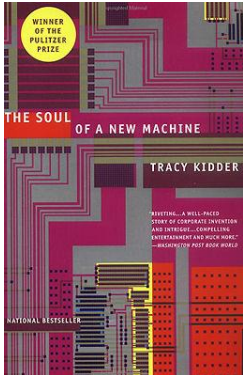
What did we cover

- Purpose for setting usability and user experience goals
- Types of usability and user experience goals
- Balance between choice of usability and user experience goals
- A set of design principles
- Adhering to the principles promotes designers' usability goals

Book (Optional Reading)



- The Design of Everyday Things - Don Norman



- The Soul of a New Machine - Tracy Kidder

Question?